

Statistics Assignment 1

Topics Covered

- Descriptive Statistics
- Probability
- Conditional Probability
- Bayes' Theorem
- Monty Hall Problem
- Law of Large Numbers
- Permutations & Combinations
- Random Variables
- Bernoulli Distribution
- Binomial Distribution
- Probability Estimation
- Central Limit Theorem
- Sampling Distribution
- Confidence Intervals (Basics)

Total Marks: 110 (+10 bonus)

Section A: Conceptual Questions (20 Marks)

Q1. Explain the difference between:

- a) Population and Sample
 - b) Parameter and Statistic
 - c) Random Variable and Random Experiment
 - d) Bernoulli Distribution and Binomial Distribution
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Q2.

A student says:

"If I toss a fair coin five times and get five heads, then the next toss is more likely to be tails."

Is the student correct? Explain your answer using probability concepts.

Q3.

Suppose the average salary of employees in a company is ₹18 lakh, while the median salary is ₹8 lakh.

What does this tell you about the salary distribution?

Q4.

Explain the Law of Large Numbers using the example of tossing a coin.

Why doesn't it guarantee exactly 50% heads after every experiment?

Q5.

Why is the Central Limit Theorem considered one of the most important theorems in statistics?

Section B: Descriptive Statistics (10 Marks)

Consider the following dataset representing marks of students:

None

42, 45, 49, 52, 52, 53, 55, 56, 58, 61, 90

Q6.

Calculate:

- Mean
 - Median
 - Mode
-

Q7.

If the value 90 is replaced by 60,

- Which statistic changes the most?
 - Which statistic changes the least?
 - Explain why.
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Section C: Probability (20 Marks)

Q8.

A bag contains

- 5 Red balls
- 3 Blue balls
- 2 Green balls

Find:

- a) Probability of selecting a Red ball
 - b) Probability of NOT selecting a Green ball
 - c) Probability of selecting either Blue or Green
-

Q9.

A die is rolled.

Find the probability of

- a) Getting an even number

- b) Getting a number greater than 4
 - c) Getting both even and greater than 4
-

Q10.

A card is drawn from a standard deck.

Find the probability of drawing

- a) A King
 - b) A Heart
 - c) A Red King
-

Q11.

Two fair coins are tossed.

Find

- a) Sample space
 - b) Probability of exactly one Head
 - c) Probability of at least one Head
-

Section D: Conditional Probability & Bayes (15 Marks)

Q12.

A company has

- 60% Male employees
- 40% Female employees

Among males, 20% know Python.

Among females, 50% know Python.

Answer:

- a) What is the probability that a randomly selected employee knows Python?
 - b) If an employee knows Python, what is the probability that the employee is female?
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Q13.

In a hospital,

1% of patients have Disease X.

The medical test has

- Sensitivity = 95%
- Specificity = 90%

If a patient's test result is positive,

What is the probability that the patient actually has the disease?

Use Bayes' theorem.

Section E: Counting (10 Marks)

Q14.

A password consists of

- 2 letters
- followed by 3 digits.

Letters and digits can repeat.

How many passwords are possible?

Q15.

A class has 12 students.

How many different committees of 4 students can be formed?

Q16.

How many ways can the letters in the word

None

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be arranged?

Section F: Random Variables (10 Marks)

Q17.

A fair coin is tossed three times.

Let

None

$X = \text{Number of Heads}$

Create the probability distribution table for X .

Find

- $P(X=0)$
 - $P(X=1)$
 - $P(X=2)$
 - $P(X=3)$
-

Q18.

Is X above

- Discrete
- Continuous

Explain.

Section G: Bernoulli & Binomial (10 Marks)

Q19.

A basketball player has an 80% chance of scoring.

Define a Bernoulli random variable.

Find

a) Mean

b) Variance

Q20.

The player shoots 8 times.

Find the probability of scoring exactly 6 baskets.

State clearly

- n
 - p
 - random variable
 - distribution used
-

Section H: Confidence Intervals & Sampling (5 Marks)

Q21.

A sample of 100 students has an average height of

None

170 cm

The population standard deviation is

None

10 cm

Construct a 95% confidence interval.

(Use $z = 1.96$)

Q22.

A student says

"A 95% confidence interval means there is a 95% probability that the true mean lies inside the interval."

Is this statement correct?

Explain.

Section I: Python Programming (10 Marks)

Use Python to answer the following.

Q23.

Generate 1,000 coin tosses.

Calculate

- Number of Heads
 - Number of Tails
 - Experimental Probability
-

Q24.

Simulate rolling two dice 10,000 times.

Estimate

None

$P(\text{sum} = 7)$

Compare it with the theoretical probability.

Bonus Challenge (10 Marks)

Q28. The Monty Hall Problem

Write a Python simulation that plays the Monty Hall game **10,000 times**.

Calculate the win percentage for:

- Always Switching
- Never Switching

Explain why the two probabilities differ.